

PROJECT DESIGN PHASE

Metro Ticket Generating System in ServiceNow

Problem–Solution Fit • Proposed Solution • Solution Architecture

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Project Name	Metro Ticket Generating System in ServiceNow

Overview

This phase turns the requirements gathered during ideation into a concrete ServiceNow design. It shows how each passenger pain point is answered by a platform capability, summarizes the proposed solution, and describes the layers and data flow that make up the Metro Ticket Generating System.

1. Problem – Solution Fit

Each recurring passenger problem is paired with the need behind it and the ServiceNow capability chosen to meet it.

Customer Problem	Observed Need	Proposed Capability	Fit / Benefit
Ticket booking is slow and manual	Quick, hassle-free digital booking	“Book A Metro Ticket” Service Catalog item	Cuts booking time and effort
Fares are hard to work out for each route	Correct fare shown before confirming	Catalog Client Script with a station-to-station fare matrix	Fare is calculated automatically
Passengers prefer different ways to pay	A choice of payment methods	UPI / Card / Others, with a UI Policy for extra details	Suits a wide range of users
A ticket must be shown at the gate	A ticket that is easy to display	Metro QR Widget on the Service Portal	QR ticket available digitally
Ticket data is scattered and inconsistent	One central place for booking data	Metro Database filled by Flow Designer	Records created automatically
Request numbers are hard to read	Simple, readable tracking	RITM Number field on the ticket record	Better traceability for staff

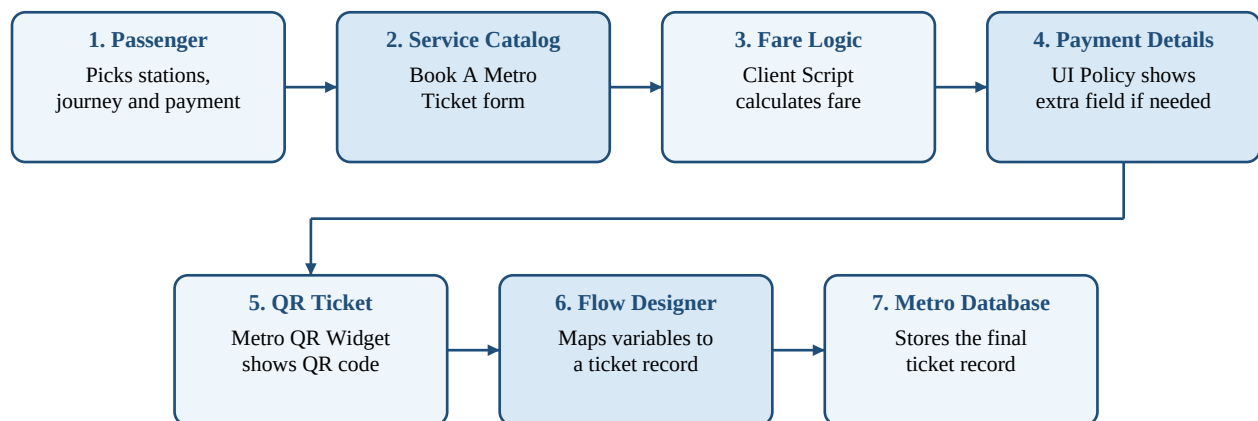
2. Proposed Solution

The table below summarizes the solution against the standard project-design parameters.

S.No.	Parameter	Description
1	Problem Statement	Metro passengers need a simple, quick, and dependable way to book tickets digitally, with the fare worked out correctly and a ticket they can show easily at the gate.
2	Idea / Solution Description	Build a Metro Ticket Generating System on ServiceNow. Passengers open the Service Portal (/sp), search for Book A Metro Ticket , and choose their source and destination stations, passenger type, and number of tickets. A Client Script calculates the fare, a UI Policy shows the extra payment field only when needed, the Metro QR Widget displays a QR ticket, and Flow Designer records the booking in the Metro Database.
3	Novelty / Uniqueness	Brings station-based fare automation, conditional payment details, QR ticket display, and ServiceNow request tracking together in a single workflow, built from standard ServiceNow features (Service Catalog, Flow Designer, widgets, and Notifications) plus a QR image URL, rather than a separate ticketing product.
4	Social Impact / Customer Satisfaction	Shorter waits at counters, less manual work for passengers and staff, support for cashless payment, and less paper because tickets are digital.
5	Business Model (Revenue Model)	Primarily a service-automation solution. In a live metro setting it would support fare collection and smoother station operations. The project brief does not define a specific revenue model.
6	Scalability of the Solution	New stations, fare rules, payment integrations, reports, notifications, and real gate-level ticket validation can be added on top of the current workflow without redesigning it.

3. Solution Architecture

The design is organized into layers, from the passenger at the top to the data and security layers underneath. The diagram shows how a booking moves through them.



Layers and Responsibilities

Layer	Component	Responsibility
User Layer	Passengers (primary); Station Managers, Metro Operations Team, IT Admins (supporting)	Passengers choose route, journey, passenger, and payment options; staff oversee operations and configuration.
Experience Layer	Service Portal / Service Catalog	Presents the Book A Metro Ticket form and the booking confirmation.
Logic Layer	Catalog Client Script	Validates the chosen route and calculates the fare.
Logic Layer	Catalog UI Policy	Shows or hides the Enter Payment Mode field.
Digital Ticket Layer	Metro QR Widget + QR image URL	Displays the QR representation of the ticket.
Automation Layer	Flow Designer (Metro Project) and Notifications	Reads catalog variables, creates the Metro Database record, and supports booking notifications.
Data Layer	Metro Station's Details (u_metro_station_details)	Holds the station master list.
Data Layer	Metro Database	Holds ticket number, RITM, route, passenger details, fare, payment mode, and QR status.
Security Layer	ServiceNow ACLs / Roles	Default table ACLs keep the custom tables out of direct reach of portal users.

Key Design Decisions

- **Station reference data:** the Metro Station's Details table (single field, Station Name — *u_station_name*) supplies the choices for Starting From and Going To.
- **Fare on the form:** the fare is calculated on the catalog form from a configured station-to-station fare matrix, so the passenger sees the amount before submitting.
- **Readable tracking:** Request Item stays a reference to *sc_req_item*, while a separate RITM Number text field stores the number in a readable form.
- **Ticket numbering:** the Metro Database uses ServiceNow auto-numbering with the prefix MTKT followed by five digits.
- **Automated records:** Flow Designer copies the catalog variables into the Metro Database and maps Requested Item → Number into RITM Number.
- **QR presentation:** the Metro QR Widget (ID *metro_qr_widget*) receives a QR image URL and shows it on the portal.
- **Access:** passengers book through the catalog only; the default ACLs created with each custom table are used, and stricter role-based rules are a future enhancement.